

ANTIFUNGAL DRUGS

Prof. Dr. Mai Abdul Alim A. Sattar

Dr. Abdulhadi Burzangi

Objectives

- Describe the mechanisms of action, pharmacokinetics and adverse effects of the major drugs used for fungal infections
- Apply the pharmacological effects to their major clinical applications

Drugs acting on fungi

Alter cell
membrane permeability

Block nucleic
acid synthesis

Disrupt
microtubule functions

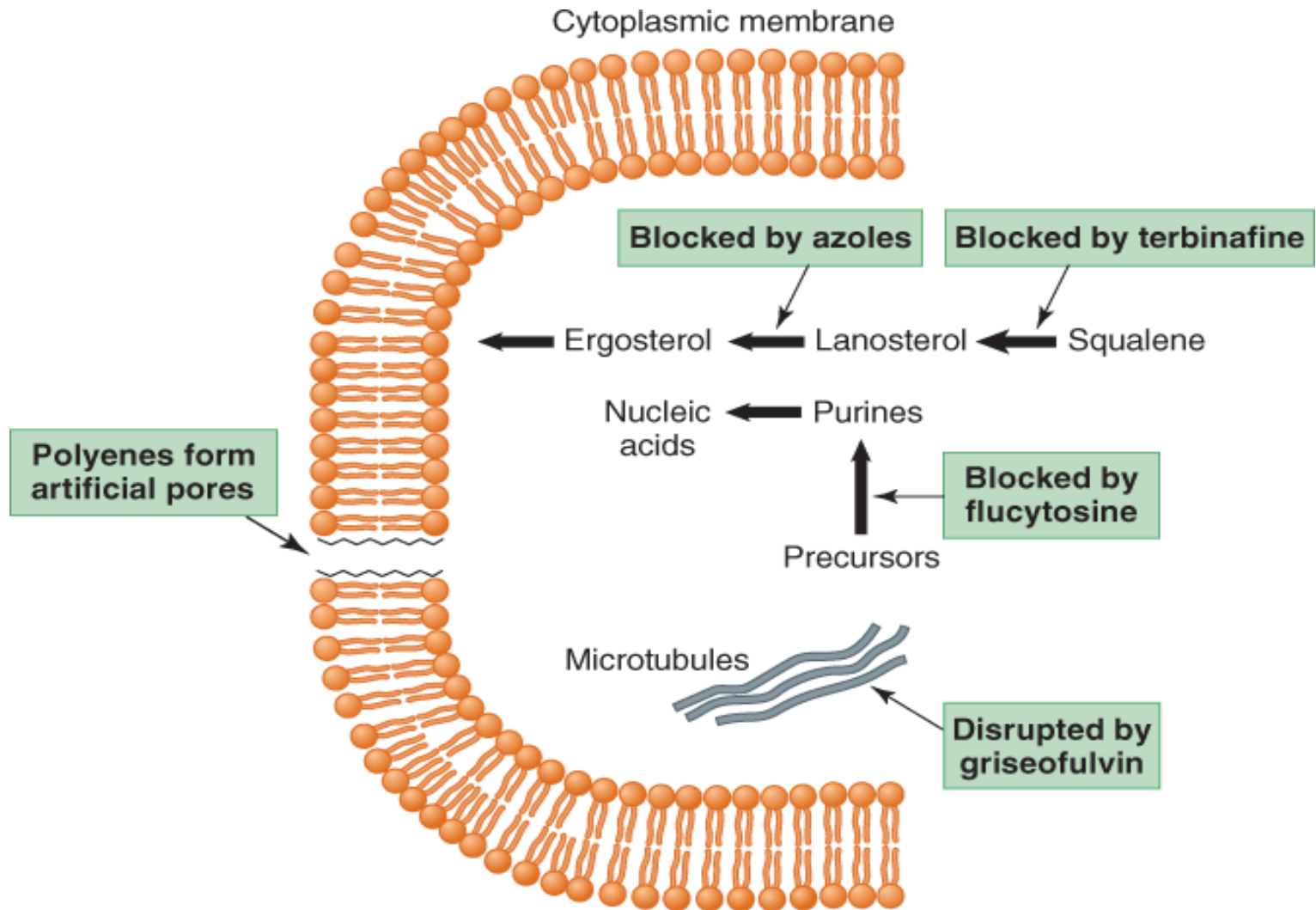
Azoles

Polyenes

Terbinafine

Flucytosine

Griseofulvin

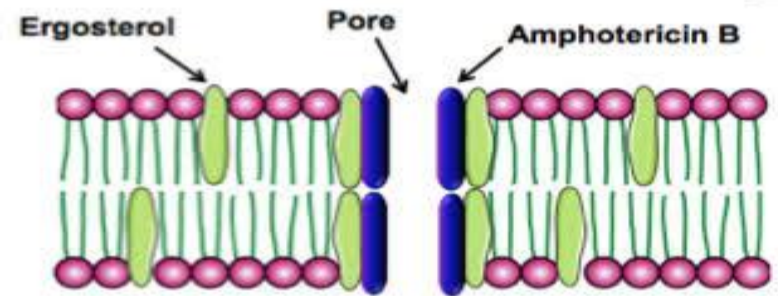


-Fungal infections have increased in incidence and severity in due mainly to advances in surgery, cancer treatment ,infection with HIV and increased use of broad spectrum antibiotics.

- **Important fungal infections are classified into:**
- **1-Superficial fungal infections:**
- **Dermatomycosis(*cutaneous mycosis*):**
- caused by dermatophytes affecting the dead keratinized structures of **skin** (stratum corneum, **hairs, nails**) → tinea pedis, tinea capitis,
- **Candidiasis:** caused by candida albicans.
- → **2-Deep fungal infections:** aspergillus, Cryptococcus or candida, blastomyces, histoplasma & coccidioides affecting deep layers of skin(subcutaneous) & mucous membranes as well as internal organs →**systemic mycosis**, e.g. pneumonias, meningitis...

I-Deep fungal infections: A- Amphotericin B

1. The drug of choice in life threatening systemic mycoses e.g. **cryptococcal meningitis** (intrathecal injection). Also in mycotic corneal ulcers and keratitis, **fungal arthritis**



Mechanism of action:

- Binds firmly & **selectively to ergosterol** (in fungal cell membrane) → formation of "**amphotericin pores**" in cell membrane → disturbance in cell membrane permeability and transport → leakage of intracellular ions & enzymes → cell death (fungicidal death).

Resistance

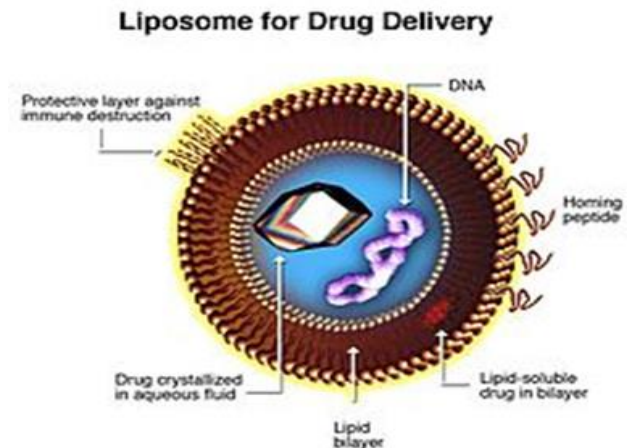
- occurs due to decreased ergosterol content.
- The drug is either ***fungicidal or fungistatic*** depending on the organism & concentration of drug.
- It affects the above mentioned organisms+ protozoal infections and leishmaniasis.
- Pharmacokinetics:
 - -This polyene is given by ***slow IV infusion***.
 - *90% is plasma protein bound.*
 - Plasma $T_{1/2}$ is approx. 2 weeks
 - Wide distributed in all tissues Except CNS.

Cont.,

May be given *intrathecally in meningitis* (Only 2-3% of blood level is reached in CSF).

Most of drug is metabolized slowly in liver.
Excretion is slow in urine, takes several days.

-Liposomal preparations (very expensive) **reduce renal & infusion toxicity.**



-Crosses placenta.

-No dose adjustment in hepatic dysfunction but with *renal dysfunction(decrease dose by 50%).*

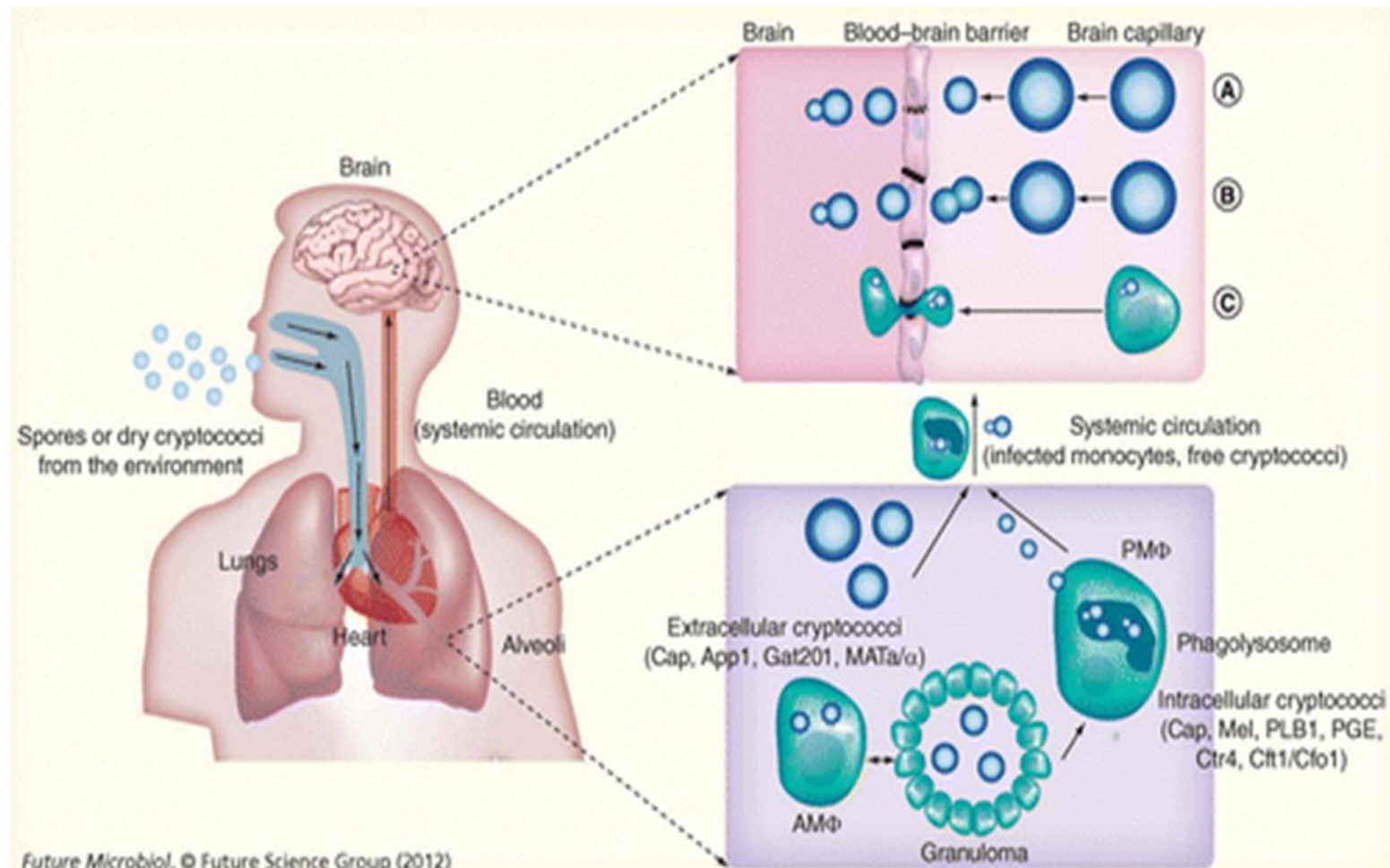
Adverse effects:

- -Of low therapeutic index. Daily dose not more than 1.5mg/kg.
- -**Small test dose** is recommended before IV infusion.
- a-Immediate:
- Rigors & fever (*prevented by pretreatment* with corticosteroids and NSAIDs), headache and vomiting.
- **Hypersensitivity and anaphylaxis.**
- Hypotension with hypokalemia which necessitates K⁺ supplement (caution must be taken in patients taking digitalis).

- **b-Delayed:**
- **Nephrotoxicity**: Dose -dependent.
- Toxicity is potentiated by **sodium depletion**(infusion of normal **saline before and after** drug infusion may decrease this toxicity), reversible, increased by nephrotoxic drugs as aminoglycosides,..
- Normocytic normochromic **anemia** by decreased erythropoietin production(↑in HIV patients taking zidovudine).
- **Thrombophlebitis** (can be diminished by adding heparin to the infusion)
- **Hepatic** dysfunction.
- **CNS stimulation** →convulsions (more if given intrathecally).

- **B-Flucytosine:**

- -**Narrow** spectrum antifungal drug e.g. *Candida albicans* or ***Cryptococcus neoformans***. Molds causing Chromoblastomycosis.



Mechanism of action:

- Converted within the fungal cell (*not the human cell*) into **5-FU**, then to **5F-dUMP** (*inhibits thymidylate synthase*) and **5F-dTMP** which *inhibit DNA & RNA synthesis* respectively.
- **5-FC & amphotericin B** are **synergistic** as amphotericin B ↑cell permeability allowing more 5-FC to enter the cell. This combination is used in the treatment of *candidiasis or cryptococcosis*.
- Combined with Itraconazole for treating chromoblastomycosis.

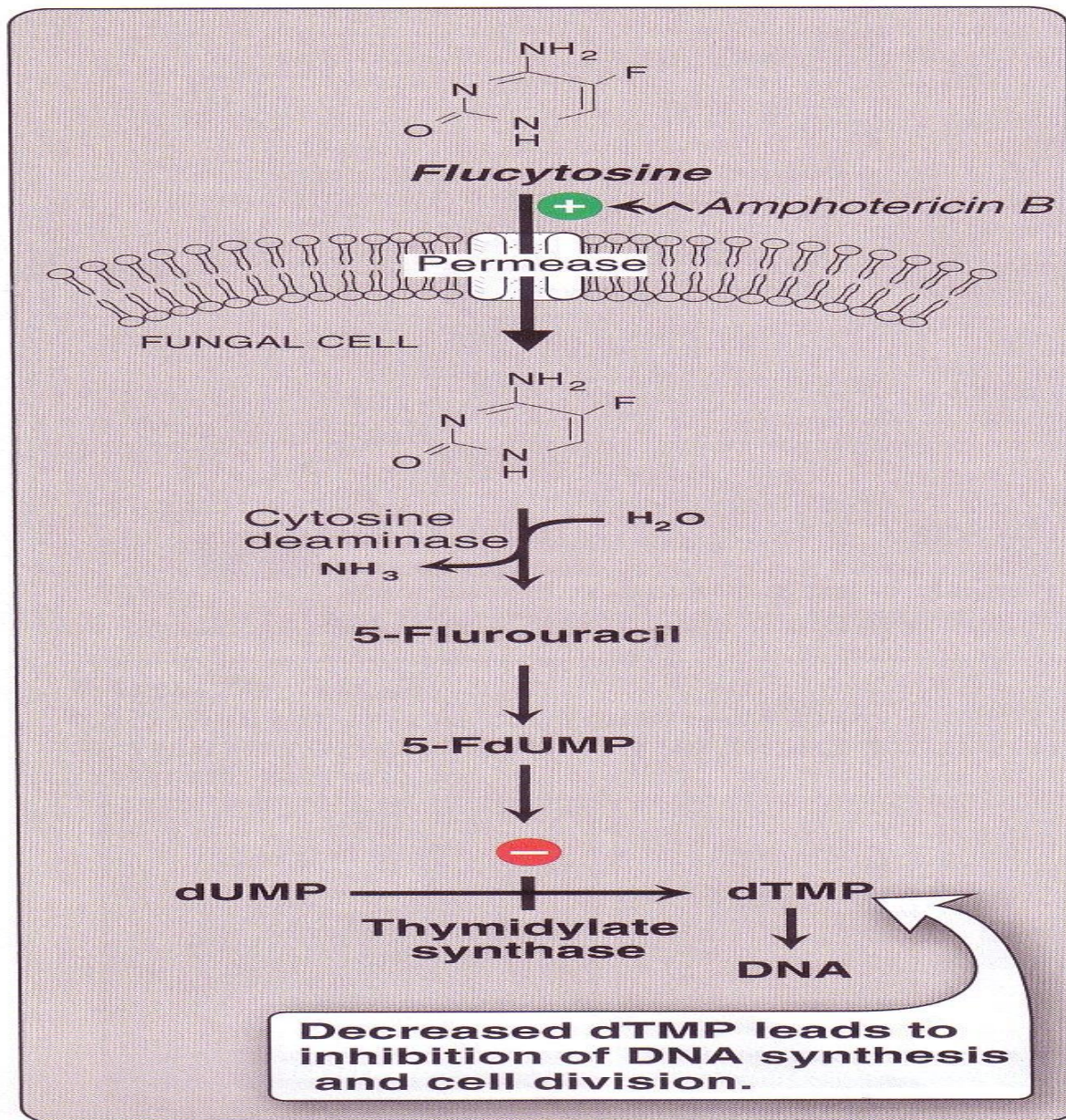


Figure 35.6

Mode of action of *flucytosine*.
5-FdUMP = 5-fluorodeoxyuridine
5'-monophosphate.

- **Pharmacokinetics:**

- Well **orally** absorbed, well distributed including CSF. Excreted unchanged via **kidney**, thus **dose adjustment** is mandatory in renal dysfunction.

- **Adverse effects:**

- (Some adverse effects are caused by 5 FU formed from 5FC).
- **Bone marrow depression**; (caution must be taken in patients taking radiation or chemotherapy).
- **Alopecia** (reversible).
- **Hepatic** dysfunction(reversibly increases serum transaminases and alkaline phosphatase).
- **GIT** disturbances, severe enterocolitis may occur.

- **C-Azoles:**

- A.Imidazoles :**

- **Ketoconazole**: systemic and topical
- **Miconazole** (**Topical**, Athlete's foot)
- **Clotrimazole** (**Topical**, vaginal yeast, oral thrush and ring worm).
- *The selectivity of imidazoles towards its target **is less than** triazoles with **more adverse effect and drug-drug interactions** than triazoles.*
- Many fungal species **are resistant**.

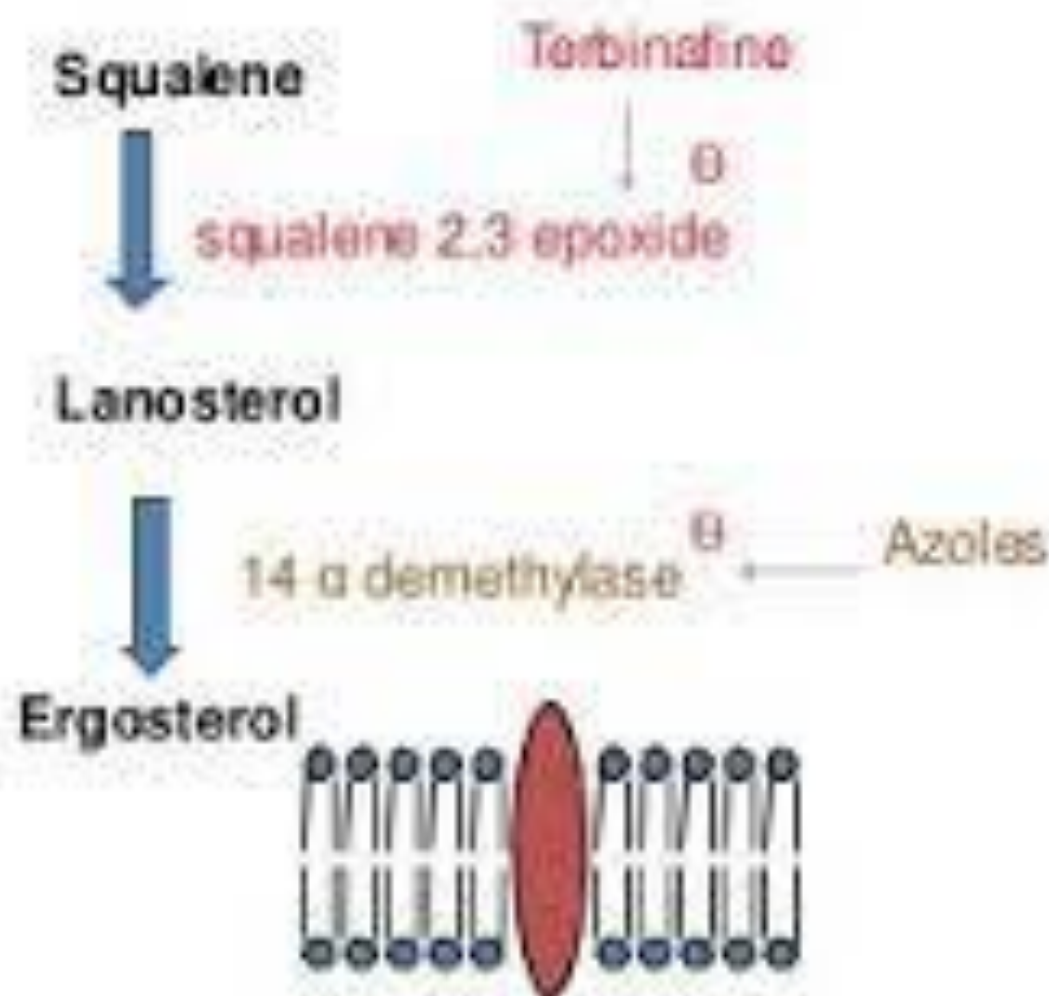
- B.Triazoles :** - Fluconazole

- Itraconazole (aspergillus)
- Voriconazole (**invasive aspergillosis**)

-

- **-Broad** spectrum fungistatic drugs (less toxic than amphotericin B).
- Mechanism of action:
- Inhibit synthesis of fungal cell membrane through inhibition of cytochrome P450-dependent C14- α demethylase enzyme which is essential for conversion of **lanosterol to ergosterol**.
- Depletion of ergosterol alters the fluidity and function of fungal cell membrane → inhibition of cell growth.
- Mutation of this enzyme can cause **resistance**, also pumping out of the drug.

Mechanism of action:



1-Ketoconazole:

- The 1st **broad** spectrum **oral** antifungal drug.
- Broad spectrum against ***superficial and systemic fungal*** infections as candida, histoplasma, blastomyces, coccidioides, **NOT** aspergillus and fungal meningitis (does **not cross BBB**).
- Used in **advanced *androgen-dependent cancer prostate*** (inhibits gonadal and adrenal steroid synthesis)
- Available **topically** (shampoo or cream) as well as tablets.

Pharmacokinetics:

- -Given ***orally requiring gastric acidity*** for dissolution and absorption. Thus, H₂ blockers, proton pump inhibitors(omeprazole) & antacids impair absorption, while acidifying agents improves absorption in patients with achlorhydria.
- -***Extensively metabolized in liver*** and excreted mainly through bile.
- Of ***low therapeutic index***.
- -Does ***not*** cross BBB

- Adverse effects:
- *Dose-dependent GIT* disturbances are the most common.
- **Liver** toxicity, the drug may accumulate in liver dysfunction (*frequent liver function tests*).
- Blocks synthesis of **adrenal & gonadal steroids** (at high doses) resulting in gynecomastia, decreased libido & fertility in males & menstrual irregularities in females.
- Pruritis & skin rash.

Cont.,

- **Drug interactions:** H2 blockers, antacids, proton pump inhibitors decrease its absorption.
- **Enzyme inhibition** → ↑levels & toxicity i.e. phenytoin, warfarin .
- Serious cardiac effects have occurred when was taken with **antihistamines** e.g. astemizole
- ***Ketoconazole and amphotericin B should not be given*** together because the decrease in ergosterol of fungal membrane reduces the fungicidal action of amphotericin.
- Should be ***avoided in pregnancy(teratogenic)***.

2-Fluconazole:

- **Broad** spectrum, given orally or IV.

The drug of choice in **Cryptococcus neoformans**, candidemia & coccidiomycosis, Esophageal and oropharyngeal candidiasis, Vaginal candidiasis.

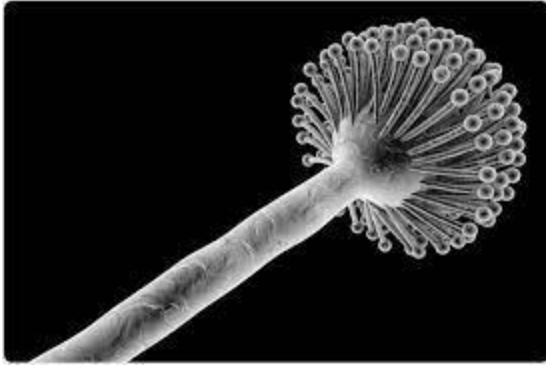
- Given prophylactically in recipients of bone marrow transplants or AIDS.
- Adverse effects:
- -GIT disturbance, skin rash, **hepatitis (rare)**, headache.
- Can **inhibit** the metabolism of other drugs.
- Better to be **avoided in pregnancy(teratogenicity)**.

- Differs from ketoconazole in:
- Its absorption does *not* depend on gastric acidity.
- Excellent *bioavailability* and *CSF penetration*.
- *Minimal* plasma protein binding activity & metabolism (*the least effect of all azoles on drug metabolism*).
- Excretion via kidney (dose adjustment in renal dysfunction).
- *Lack of endocrine* adverse effects.

3-Voriconazole

- a broad-spectrum antifungal agent. It is available for both IV and oral administration.
- Voriconazole is the treatment of choice for *invasive aspergillosis* and has replaced amphotericin B as for this indication.
- **Adverse effects:**
 - Similar to those of the other azoles.
 - High trough concentrations are associated with **visual** and auditory hallucinations.

invasive pulmonary aspergillosis



	KETOCONAZOLE	FLUCONAZOLE	VORICONAZOLE	POSACONAZOLE
SPECTRUM	Narrow	Expanded	Expanded	Expanded
ROUTE(S) OF ADMINISTRATION	Oral	Oral, IV	Oral, IV	Oral
$t_{1/2}$ (HOURS)	6-9	30	6-24	20-66
CSF PENETRATION	No	Yes	Yes	Yes
RENAL EXCRETION	No	Yes	No	No
INTERACTION WITH OTHER DRUGS	Frequent	Occasional	Frequent	Frequent
INHIBITION OF MAMMALIAN STEROL SYNTHESIS	Dose-dependent inhibitory effect	No inhibition	No inhibition	No inhibition

Figure 35.12 Summary of some azole fungistatic drugs.

- D. Echinocandins Caspofungin:
- Caspofungin has activity against ***Aspergillus and most Candida*** species, including those species ***resistant to azoles***.
- Caspofungin should not be administered with cyclosporine, due to the possible increase of serum concentration of caspofungin caused by cyclosporine.
- the solution should be administered by slow ***intravenous infusion*** over approximately 1 hour.

- **II- Drugs for cutaneous mycotic infections**

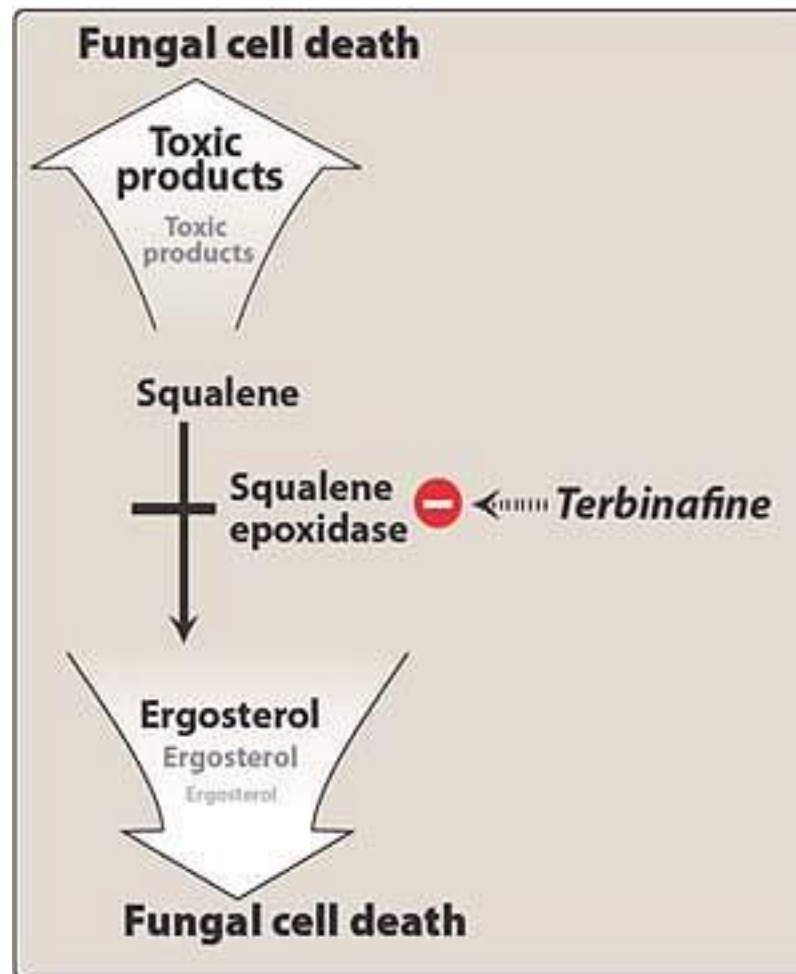
- **1-Systemic antifungal drugs:**

Fungi which cause superficial skin infections are called dermatophytes.

- **A-Terbinafine**

- Of choice for treating **dermatophytes** (especially those of nails; **onychomycosis**) and **Candida albicans**.
- *-Better tolerated, shorter duration of therapy(3 months) and more effective than either griseofulvin & itraconazole.*

- **Mechanism of action:** Selectively **inhibits fungal squalene epoxidase** enzyme, which is involved in ergosterol synthesis from squalene in fungal cell membrane → ***accumulation of toxic squalene*** within the cell → cell death.



- Active orally, deposited in skin, nails & fat. It accumulates in breast milk (***contraindicated in lactating mothers***).
- *Dose should be reduced in renal & hepatic dysfunction*(it is extensively metabolized before renal excretion).
- But does not affect the metabolism of other drugs
- **Adverse effects:**
- GIT disturbances are the most common, headache, pruritis, rash, joint & muscle pain and rarely hepatitis, taste and visual disturbances.

- Topical antifungal drugs
- A-Nystatin
- Similar to amphotericin B.
- Used topically for treatment of candida infection (*oral thrush*) because of *its systemic toxicity*.
- It is not absorbed systemically → acts **locally** after oral administration for treatment of *oral candidiasis*.



B-Azoles

- **Miconazole, clotrimazole** (rarely used systemically because of *sever toxicity*) are used in: Oral thrush, Vulvovaginal candidiasis, Tinea cruris, Tinea corporis and Tinea pedis

(athlete's foot)



- **Ketoconazole** (effective in seborrheic dermatitis).
- **Miconazole** is a potent **enzyme inhibitor**, resulted in bleeding in patients taking warfarin even if applied topically.
- **C.Tolnaftate**: **not effective** against candida.
- Used to treat tinea pedis, cruris and corporis.

DRUG SUMMARY TABLE: ANTIFUNGAL DRUGS

Drug/Drug Class	Mechanism of Action	Clinical Applications	Pharmacokinetics & Interactions	Toxicities
Amphotericin B	Binds to ergosterol in fungal cell membranes, forming "leaky pores"	Candidemia and infections caused by Aspergillus, Blastomyces, Cryptococcus, Histoplasma, Mucor, etc	Multiple forms, IV for systemic infections (liposomal forms less nephrotoxic); topical for ocular/bladder infections	Nephrotoxicity is dose-limiting, additive with other nephrotoxic drugs; infusion reactions (chills, fever, muscle spasms, hypotension)
Azoles Ketoconazole Fluconazole Itraconazole Posaconazole Voriconazole	Inhibit fungal P450-dependent enzymes blocking ergosterol synthesis; resistance can occur with long-term use	Aspergillosis (voriconazole); blastomycosis (itraconazole, fluconazole); mucormycosis (posaconazole); alternative drugs in candidemia and infections caused by Aspergillus, Blastomyces, Cryptococcus, andHistoplasma	Various topical and oral forms for dermatophytoses Oral, parenteral forms for mycoses (fluconazole, itraconazole, posaconazole, voriconazole) Most azoles undergo hepatic metabolism; fluconazole eliminated in urine unchanged	Ketoconazole rarely used in systemic fungal infections owing to its inhibition of hepatic and adrenal P450s; other azoles are less toxic, but may cause GI upsets and rash; voriconazole causes visual disturbances and is class D re pregnancy risk
Echinocandins Caspofungin Micafungin Anidulafungin	Inhibit -glucan synthase decreasing fungal cell wall synthesis	Treatment of candidemia; caspofungin is also used as "salvage" therapy in apergillosis	IV forms; micafungin increases levels of nifedipine and cyclosporine	Gastrointestinal (GI) distress, flushing from histamine release
Flucytosine	Inhibits DNA and RNA polymerases	Synergistic with amphotericin B in candidemia and cryptococcal infections	Oral; enters cerebrospinal fluid; renal elimination	Bone marrow suppression
Terbinafine	Inhibits epoxidation of squalene	Mucocutaneous fungal infections; accumulates in keratin	Oral; long duration of action (weeks)	GI upsets, headache